METALLURGY →

① Seven metals of antiquity → gold, Cu, Ag, Pb, Sn, Fe, Hg

② Ruby → Cr
Sapphire → Co
[As impurity]

③ 2nd most abundant → Fe

④ Kaolinite (a form of clay) → $[Al_2(OH)_4Si_2O_5]$

⑤ Collectors → pine oil, fatty acids, Xanthates
Stabilisers → aniline, Cresol
→ Froth Floatation

⑥ a) Distillation → Zn, Hg

b) Liquation → Sn

c) Electrolytic Refining → Cu, Zn
↳ Anode mud – Se, Te, Ag, Au, Pt, Sb (antimony)

d) zone – Refining → Ge, Si, B, Ga, In

e) vapour phase Refining →

a) Mond's Process – Ni

$$Ni + 4CO \xrightarrow{330-350K} Ni(CO)_4$$
$$Ni(CO)_4 \xrightarrow{450-470K} Ni + 4CO$$

b) Van – Arkel → Zr, Ti

$$Zr + 2I_2 \longrightarrow ZrI_4$$
$$ZrI_4 \xrightarrow{1800K} Zr + 2I_2$$ (in tungsten filament)

⑦ Brass → Cu / Zn
Bronze → Cu / Sn
Coinage alloy → Cu / Ni
German Silver → Cu / Zn / Ni

⑧ Zn → galvanising iron